The Support of the Character of a Supercuspidal Representation

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Note (*) by Mr. Pierre Deligne, communicated by Mr. Jean-Pierre Serre.

The character of a supercuspidal representation of a semisimple group over a local field is supported on the union of the compact open subgroups.

1 Review of monogenic groups

Let V be a finite-dimensional vector space over a field k, and let $g \in \mathrm{GL}(V)$. If k is perfect, the Zariski closure $\langle g \rangle$ of the group generated by g is the product of a multiplicative group $\langle g \rangle_m$ and a unipotent group $\langle g \rangle_{\mathrm{un}}$. If $g = su = us$ is the Jordan decomposition of g into its semisimple and unipotent parts, then $\langle g \rangle_m = \langle s \rangle$ and $\langle g \rangle_{\mathrm{un}} = \langle u \rangle$. Without any hypothesis on k, define the subgroup $\langle g \rangle_m$ of $\langle g \rangle$ to be the Zariski closure of the union of the subgroups of $\langle g \rangle$ that are kernels of multiplication by n, for n relatively prime to the characteristic exponent of k. This definition is compatible with extension of scalars and is equivalent to the preceding one when k is perfect. Over an algebraic closure $\bar{k}$ of k, the character group of $\langle g \rangle_m$ is the subgroup X of $\bar{k}^*$ generated by the eigenvalues of g, and

$$s \in \langle g \rangle_m(\bar{k}) = \mathrm{Hom}(X, \bar{k}^*)$$

corresponds to the identity inclusion of X.

Suppose that k is complete with respect to a valuation and has uniformizer π. Let v be the valuation of $\bar{k}$, with values in $\mathbf{Q}$, normalized by $v(\pi) = 1$, and let $n > 0$ be an integer such that $nv(X) \subset \mathbf{Z}$. The homomorphism $nv(x) : X \to \mathbf{Z}$ transposes to $m_g : \mathbf{G}_m \to \langle g \rangle_m$.

For a linear algebraic group G over k and $g \in G(k)$, the preceding constructions may be repeated after choosing an embedding $G \subset \mathrm{GL}(V)$. The homomorphism $m_g : \mathbf{G}_m \to \langle g \rangle_m \subset G$ does not depend on the chosen embedding (for fixed n). For every linear representation W of G, it defines an action of $\mathbf{G}_m$ on W, that is, a grading of W (SGA 3 I 4.7.3). The degree-d part is the sum of the generalized eigenspaces of g corresponding to eigenvalues $\alpha \in \bar{k}$ such that $nv(\alpha) = d$.

Suppose that G is reductive. Every homomorphism m from $\mathbf{G}_m$ to G then defines two opposite parabolic subgroups of G, the parabolics contracted and dilated by m. We write P_g^+ for the parabolic contracted by m_g (also called contracted by g). It has the following equivalent characterizations:

(a) An element $x \in G(\bar{k})$ lies in P_g^+ (respectively, in its unipotent radical U_g^+) if and only if the morphism

$$\lambda \longmapsto m_g(\lambda) x m_g(\lambda)^{-1} : \mathbf{G}_m \longrightarrow G$$

extends to $\mathbf{G}_a$ [respectively, and $\lim_{\lambda \to 0} m_g(\lambda) x m_g(\lambda)^{-1} = e$] — equivalently, if and only if $\lim_{n\to\infty} g^n x g^{-n}$ exists (respectively, equals e).

(b) $\mathrm{Lie}\, P_g^+$ (respectively, $\mathrm{Lie}\, U_g^+$) is the sum, in $\mathrm{Lie}\, G$, of the generalized eigenspaces of $\mathrm{ad}\, g$ corresponding to eigenvalues of valuation ≥ 0 (respectively, > 0).

Set $P_g^- = P_{g^{-1}}^+$ [the parabolic dilated by g and contracted by $m_{g^{-1}} = (m_g)^{-1}$], and denote its unipotent radical by U_g^-. The parabolics P_g^+ and P_g^- are opposite, and we set $M_g = P_g^+ \cap P_g^-$.

2 Statement and proof of the theorem

Assume that k is locally compact and that G is reductive. Let τ denote the projection of G onto its adjoint group G^{ad}. An admissible $(\mathscr{H},\rho)$ representation of $G(k)$ is *supercuspidal* if its matrix coefficients are supported in inverse images of compact subsets of $G^{\mathrm{ad}}(k)$.

If K is a compact subgroup of $G(k)$, let dk denote the normalized Haar measure on K, and let $[K]$ denote its image in the algebra of compactly supported distributions on $G(k)$. The endomorphism

$$\rho(K) := \rho([K])$$

is a projector from $\mathscr{H}$ onto the subspace $\mathscr{H}^K$ of K-invariants. Equivalently, supercuspidality of ρ means that, for every compact open subgroup K,

$$\rho(K)\rho(g)\rho(K) = 0$$

whenever $\tau(g) \in G^{\mathrm{ad}}$ lies outside a suitable compact subset (depending on K). Let G_c^{ad} denote the union of the compact subgroups (or, equivalently, the compact open subgroups) of $G^{\mathrm{ad}}(k)$.

Theorem — *If $(\mathscr{H},\rho)$ is an admissible supercuspidal representation of $G(k)$, its character*(1) *is supported in $\tau^{-1}(G_c^{\mathrm{ad}})$.*

Choose a system of local coordinates near e, that is, a lattice $L \subset \mathrm{Lie}(G)$ and an analytic isomorphism α from L onto a neighborhood of e, such that

(1) $\alpha(0) = e$, and $d\alpha$ at 0 is the identity of $\mathrm{Lie}(G)$.

Let $g \in G(k)$. Choose L so that

(2) $L = (L \cap \mathrm{Lie}\, U_g^+) \oplus (L \cap \mathrm{Lie}\, M_g) \oplus (L \cap \mathrm{Lie}\, U_g^-)$. Denote this decomposition by $L^+ \oplus L^0 \oplus L^-$.
(3) $\mathrm{ad}\, g(L^+) \subset L^+$, $\mathrm{ad}\, g(L^0) = L^0$, and $\mathrm{ad}\, g^{-1}(L^-) \subset L^-$.

Since P_g^+ and P_g^- are stable under $\mathrm{ad}\, g$, condition (3) may be rewritten as

(3') $\mathrm{ad}\, g(L \cap \mathrm{Lie}\, P_g^+) \subset L$ and $\mathrm{ad}\, g^{-1}(L \cap \mathrm{Lie}\, P_g^-) \subset L$.

Let a subscript i denote intersection with $G_i = \alpha(\pi^i L)$. Under hypotheses (2) and (3), there exists i_0 such that

(4) for $i \geq i_0$, G_i is a subgroup of $G(k)$, $G_i = U_{gi}^+ \cdot M_{gi} \cdot U_{gi}^-$, and
(5) $\mathrm{Int}\, g(U_{gi}^+) \subset U_{gi}^+$, $\mathrm{Int}\, g(M_{gi}) = M_{gi}$, and $\mathrm{Int}\, g^{-1}(U_{gi}^-) \subset U_{gi}^-$.

This follows from the decomposition

$$U_g^+ \times M_g \times U_g^- \hookrightarrow G : (u^+, m, u^-) \longmapsto u^+ m u^-$$

of a neighborhood of e.

Moreover, for i_0 sufficiently large, there is a neighborhood V of g such that (2)–(5) remain valid when g is replaced by $g' \in V$. For (2) and (3'), one uses the continuous dependence of P_g^+ and $\mathrm{ad}\, g$ on g; for (4) and (5), one uses the analytic dependence of m_g, and hence of $P_g^{\pm}$, on g. Thus, in the decomposition $x = u^+ m u^-$, the components u^+, m, and u^- are analytic functions of the pair (x, g).

Lemma — *If conditions (1)–(5) hold, $i \geq i_0$, and $\tau(g) \notin G_c^{\mathrm{ad}}$, then $\rho(G_i)\rho(g)\rho(G_i)$ is nilpotent.*

For the decomposition $U_g^+ \times M_g \times U_g^- \to G$ of the big cell of G, Haar measures satisfy

$$dg = \lambda(m)\, du^+\, dm\, du^-,$$

where λ is a character of M_g. On every compact subgroup of M_g, $\lambda = 1$; hence

$$[G_i] = [U_{gi}^+] \star [M_{gi}] \star [U_{gi}^-].$$

We prove by induction on N that

$$([G_i]g[G_i])^N = [G_i]g^N[G_i]$$

in the algebra of compactly supported distributions on G (we write g for δ_g). Indeed,

$$\begin{aligned}([G_i]g^N[G_i])([G_i]g[G_i]) &= [G_i]g^N[G_i]g[G_i] \\ &= [G_i]g^N[U_{gi}^+][M_{gi}][U_{gi}^-]g[G_i] \\ &= [G_i][g^N U_{gi}^+ g^{-N}][g^N M_{gi} g^{-N}]g^N g[g^{-1}U_{gi}^- g][G_i] \\ &= [G_i]g^{N+1}[G_i].\end{aligned}$$

If $\tau(g) \notin G_c^{\mathrm{ad}}$, the set of $\tau(g^N)$, $N \geq 1$, is not relatively compact. Since ρ is supercuspidal, there is therefore an N such that

$$(\rho(G_i)\rho(g)\rho(G_i))^N = \rho(G_i)\rho(g^N)\rho(G_i) = 0.$$

Proof of the theorem — For $g \in G(k)$ such that $\tau(g) \notin G_c^{\mathrm{ad}}$, choose L, α, i_0, and V as above, with $gG_{i_0} \subset V$. Let Θ denote the character of ρ. For every $g' \in gG_i$ and every $i \geq i_0$, we then have

$$\Theta(g'[G_i]) = \Theta([G_i]g'[G_i]) = \mathrm{Tr}(\rho(G_i)\rho(g')\rho(G_i)) = 0$$

by the lemma. Since the subgroups G_i form a fundamental system of neighborhoods of 0, Θ vanishes on gG_i, and its support does not contain g.

Remark 1 — The same theorem, with the same proof, holds for an analytic group that is a finite étale covering of an open subgroup of G. The hypothesis becomes "the image in G^{ad} of the support of the matrix coefficients is relatively compact"; the conclusion becomes "the image in G^{ad} of the support of Θ lies in G_c^{ad}".

Remark 2 — Let p be the residual characteristic of k. The proof makes essential use of the Haar measures $[K]$ only for K small, and in particular for a p-group. The theorem and its proof therefore hold for representations with coefficients in any field of characteristic different from p.

Source notes

(*) Session of 24 May 1976.

(1) Once a Haar measure dg on G has been chosen, the character Θ of ρ is the distribution on G whose value on a test function φ is

$$\mathrm{Tr}\left(\int \varphi(g)\rho(g)\,dg\right).$$

More intrinsically, it is the generalized function [the linear form on the space of measures $\varphi(g)\,dg$, where $\varphi(g)$ is locally constant with compact support and dg is a Haar measure] given by this same formula.

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